Math 110, Fall 2026 Practice Midterm Prof. Hammond

Preparation for Monday, September 28, 2026

NAME: _____ **SID:** _____

LAST NAME, OTHER NAMES (IN BLOCK CAPITALS)

The suggested time is 50 minutes. There are 4 questions worth a total of 40 points, with points per part indicated in parentheses. Try the exam before consulting the solutions.

NOTATION

$\mathcal{P}_n(\mathbb{R})$ denotes the real vector space of polynomials of degree at most n. For a linear map T, its null space and range are denoted by $\operatorname{null} T$ and $\operatorname{range} T$. All bases are ordered when coordinates or matrices are requested.

INSTRUCTIONS

1. Work under exam conditions if possible, and write only within the black borders.
2. Give concise calculations or reasoning as well as the final answer.
3. You may quote a result from the text or course materials if you state it clearly, unless the problem asks you to prove it.
4. Extra space for each question is provided on the back of the preceding page. Use that page only for the indicated question.
5. The back of the final page may be used for rough work on any question.

Extra space for QUESTION 1 ONLY

Name: _____

1. A direct sum in $\mathbb{R}^4$. Let

$$u_1 = (1, 1, 0, 0), \qquad u_2 = (0, 1, 1, 0),$$

and let

$$w_1 = (0, 0, 1, 1), \qquad w_2 = (1, 0, 0, -1).$$

Define

$$U = \text{span}(u_1, u_2), \qquad W = \text{span}(w_1, w_2).$$

a) (6) Prove that $\mathbb{R}^4 = U \oplus W$.

b) (2) Write $(2, 1, 3, 0)$ in the form $u + w$ with $u \in U$ and $w \in W$.

Extra space for QUESTION 2 ONLY

Name: _____

2. A left inverse. Let V and W be finite-dimensional and let $T : V \to W$ be an injective linear map.

a) (4) Choose a basis $v_1, \ldots, v_n$ of V. Prove that

$$Tv_1, \ldots, Tv_n$$

is linearly independent in W.

b) (6) Prove that there exists a linear map $S : W \to V$ such that

$$ST = I_V.$$

Extra space for QUESTION 3 ONLY

Name: _____

3. A matrix of a linear map. Define $T : \mathbb{R}^3 \to \mathcal{P}_2(\mathbb{R})$ by

$$T(a, b, c) = (a + b) + (b + c)x + (a + c)x^2.$$

Use the basis

$$\mathcal{B} = \big((1, 0, 0), (1, 1, 0), (1, 1, 1)\big)$$

of the domain and the basis $\mathcal{C} = (1, x, x^2)$ of the codomain.

a) (6) Find $\mathcal{M}(T)$ with respect to the basis $\mathcal{B}$ of the domain and the basis $\mathcal{C}$ of the codomain.

b) (2) Prove that T is an isomorphism.

Extra space for QUESTION 4 ONLY

Name: _____

4. The inverse image of a subspace. Let V be finite-dimensional, let $T : V \to W$ be linear, and let U be a subspace of W. Define

$$T^{-1}(U) = \{v \in V : Tv \in U\}.$$

You may use without proof that $T^{-1}(U)$ is a subspace of V. Consider the map

$$S : T^{-1}(U) \longrightarrow U \cap \operatorname{range} T, \qquad S(v) = Tv.$$

a) (4) Prove that $S(v) \in U \cap \operatorname{range} T$ for every $v \in T^{-1}(U)$. Then prove that S is linear and surjective.

b) (2) Prove that $\operatorname{null} S = \operatorname{null} T$.

c) (4) Apply the fundamental theorem of linear maps to S to prove that

$$\dim T^{-1}(U) = \dim \operatorname{null} T + \dim\big(U \cap \operatorname{range} T\big).$$

d) (4) Let $D : \mathcal{P}_4(\mathbb{R}) \to \mathcal{P}_3(\mathbb{R})$ be differentiation, and let

$$U = \operatorname{span}(1, x^2) \subseteq \mathcal{P}_3(\mathbb{R}).$$

Use part (c) to determine $\dim D^{-1}(U)$, and then give a basis of $D^{-1}(U)$.

This blank page can be used for rough work on any question.